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Electronic vaporization device having a heat shield with airflow channel

Abstract

An electronic vaporization device includes: a housing; a vaporization assembly arranged in the housing, the vaporization assembly having a vaporization base; a heat shield arranged in the vaporization base; and a heating pot arranged in the heat shield. An airflow channel for ambient air to enter the heating pot is formed on an outer surface of the heat shield.

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION

(1) Priority is claimed to Chinese Patent Application No. 202123174215.X, filed on Dec. 15, 2021, the entire disclosure of which is hereby incorporated by reference herein.

FIELD

(2) The present disclosure relates to the field of vaporization, and more specifically, to an electronic vaporization device.

BACKGROUND

(3) In the related art, when a baking-type electronic vaporization device operates, an aerosol-

forming substrate (for example, a vaporizable medium such as plant grass, tobacco paste, or shredded tobacco) filled in a heating pot is heated to generate vapor. A user sucks on a suction nozzle, inhales external air into the electronic vaporization device, and the external air passes through the heating pot, thereby driving the vapor into the mouth of the user.

(4) Because the air inhaled into the electronic vaporization device is ambient air (normal temperature), when the air comes into contact with the vapor in the heating pot, the air causes the vapor/heating pot to cool down. Or in more serious cases, the air causes the vapor to condense, affecting the taste of inhaling or leading to low efficiency of heating and vaporization.

(5) In addition, in the related art, when some baking-type electronic vaporization devices operate, the temperature of the surface of the housing is very high, which makes the user feel hot and brings a poor user experience.

SUMMARY

(6) In an embodiment, the present invention provides an electronic vaporization device, comprising: a housing; a vaporization assembly arranged in the housing, the vaporization assembly comprising a vaporization base; a heat shield arranged in the vaporization base; and a heating pot arranged in the heat shield, wherein an airflow channel for ambient air to enter the heating pot is formed on an outer surface of the heat shield.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

(2) FIG. 1 is a three-dimensional schematic structural diagram of an electronic vaporization device according to some embodiments of the present disclosure.

(3) FIG. 2 is a schematic diagram of a three-dimensional exploded structure of the electronic vaporization device shown in FIG. 1.

(4) FIG. 3 is a schematic cross-sectional structural diagram in an A-A direction of the electronic vaporization device shown in FIG. 1.

(5) FIG. 4 is a schematic cross-sectional structural diagram in an A-A direction of the electronic vaporization device in a disassembled status shown in FIG. 1.

(6) FIG. 5 is a schematic diagram of a three-dimensional exploded structure of the electronic vaporization device shown in FIG. 1 with a suction nozzle assembly removed.

(7) FIG. 6 is a schematic cross-sectional structural diagram in a B-B direction of the vaporization assembly shown in FIG. 5.

(8) FIG. 7 is a schematic diagram of a three-dimensional exploded structure of the vaporization assembly shown in FIG. 5.

(9) FIG. 8 is a schematic cross-sectional structural diagram of the vaporization assembly in a disassembled status shown in FIG. 7.

(10) FIG. 9 is a schematic diagram of a three-dimensional exploded structure of the vaporization assembly shown in FIG. 7 from another perspective.

(11) FIG. 10 is a schematic cross-sectional structural diagram in a B-B direction of the electronic vaporization device shown in FIG. 1 with a suction nozzle assembly removed.

DETAILED DESCRIPTION

(12) In an embodiment, the present invention provides an improved electronic vaporization device.

(13) In an embodiment, the present invention provides an electronic vaporization device, including

a housing and a vaporization assembly arranged in the housing, where the vaporization assembly includes a vaporization base, a heat shield arranged in the vaporization base, and a heating pot arranged in the heat shield, and an airflow channel for ambient air to enter the heating pot is formed on an outer surface of the heat shield.

(14) In some embodiments, the airflow channel is distributed over most areas of the outer surface of the heat shield.

(15) In some embodiments, the airflow channel extends from an upper end to a lower end of the heat shield.

(16) In some embodiments, the airflow channel is distributed in an S shape on the outer surface of the heat shield.

(17) In some embodiments, the heat shield includes a cylindrical first side wall, the airflow channel includes a plurality of C-shaped air guide grooves spaced apart in parallel in a longitudinal direction of an outer surface of the first side wall, and the air guide grooves are connected end to end.

(18) In some embodiments, the first side wall includes a main body portion with a C-shaped cross section and a protruding portion with a U-shaped cross section that is mouth-to-mouth combined with the main body portion, and the plurality of C-shaped air guide grooves are formed in a longitudinal direction of an outer surface of the main body portion.

(19) In some embodiments, the vaporization assembly includes a temperature sensing element, the heating pot includes a cylindrical second side wall, and the second side wall is axially arranged in the first side wall; and the temperature sensing element is arranged between the second side wall and the first side wall, and is arranged in a space defined by the protruding portion.

(20) In some embodiments, the vaporization base includes a cylindrical third side wall, and a first air inlet hole that is in communication with the airflow channel is provided on the third side wall and at a position close to an upper end of the third side wall.

(21) In some embodiments, the housing is provided with a second air inlet hole that is in communication with the first air inlet hole.

(22) In some embodiments, an outer surface of the third side wall is provided with a seal base surrounding the first air inlet hole; and the vaporization assembly includes a first seal ring embedded in the seal base, and the first seal ring causes the first air inlet hole to be in communication with the second air inlet hole in a sealing and air guiding manner.

(23) In some embodiments, the vaporization base includes an annular plate-shaped top wall that is integrally connected to the upper end of the third side wall and a plate-shaped bottom wall that is in snap-fit connection to a lower end of the third side wall, and the upper end and the lower end of the heat shield respectively abut against the top wall and the bottom wall.

(24) In some embodiments, the heat shield includes a cylindrical first side wall and a first bottom wall combined a lower end of the first side wall, and an upper surface of the first bottom wall is provided with a thermal insulation protrusion configured to support the heating pot.

(25) In some embodiments, the heat shield includes a cylindrical first side wall, and the heating pot includes a cylindrical second side wall; and the second side wall is axially arranged in the first side wall, and a spacing is formed between an outer surface of the second side wall and an inner surface of the first side wall.

(26) In some embodiments, an outer surface of the heating pot is provided with a heating circuit, and an inner surface of the heating pot is provided with an infrared radiation coating layer.

(27) In some embodiments, the electronic vaporization device further includes a suction nozzle assembly, where the suction nozzle assembly is detachably connected to the housing and is connected to the heating pot in an air guiding manner.

(28) In some embodiments, the vaporization assembly is detachably connected to the housing.

(29) In some embodiments, the vaporization base includes a cylindrical third side wall and a plurality of elongated engagement protrusions with a wedge-shaped cross section, each of the

engagement protrusion includes a helically inclined top surface, and the engagement protrusions are spaced apart in a circumferential direction of an outer wall surface of the third side wall that is close to an upper end of the third side wall, so as to be detachably and rotatably connected to the housing.

(30) In some embodiments, the housing includes a holder, the holder includes an upper end cap, a plurality of arc-shaped buckles are protruded from a lower side of the upper end cap, and the buckles encircle to be respectively rotatably connected to the plurality of engagement protrusions.

(31) The beneficial effects of the present disclosure are: because the outer surface of the heat shield is provided with an airflow channel for ambient air to enter the heating pot, on the one hand, the heat shield can be effectively cooled to reduce the heat transferred to the outer surface of the housing; and on the other hand, when the heated air enters the heating pot again, the condensation phenomenon in the heating pot can be reduced.

(32) To provide a clearer understanding of the technical features, objectives, and effects of the present disclosure, specific implementations of the present disclosure are described with reference to the accompanying drawings.

(33) FIG. 1 to FIG. 4 show an electronic vaporization device **1** according to some embodiments of the present disclosure. The electronic vaporization device **1** can be configured to heat an aerosol-generation substrate such as plant grass, tobacco paste, or shredded tobacco, so as to generate an aerosol for a user to inhale. As shown in the figure, the electronic vaporization device **1** can be in a flat cylindrical shape, which can include a flat cylindrical housing **10** with a racetrack-shaped cross-section, a cylindrical vaporization assembly **20** arranged in the housing **10**, and a suction nozzle assembly **30** that is detachably connected to one end of the housing **10** in an axial direction and is connected to the vaporization assembly **20** in an air guiding manner. The housing **10** is configured to provide a carrier for components therein and to protect the components arranged therein. The vaporization assembly **20** is configured to accommodate and heat the aerosol-generation substrate to generate the aerosol. The suction nozzle assembly **30** is configured to discharge a mixture of the aerosol and air generated by the vaporization assembly **20** for the user to inhale. It can be understood that the housing **10** is not limited to be a flat cylindrical shape, and can also be a cylindrical shape, a square cylindrical shape, or an irregular shape.

(34) In some embodiments, the electronic vaporization device **1** may further include a main control board **40**, a microphone assembly **50**, and a power source device **60**, and the main control board **40** is configured to control the operation of the entire electronic vaporization device **1**. The microphone assembly **50** is electrically connected to the main control board **40** and is connected to the vaporization assembly **20** in an air guiding manner, so as to inhale airflow in the vaporization assembly **20**, and to be connected to the power source device **60**. The power source device **60** is configured to supply electric power to electronic components such as the vaporization assembly **20**, the main control board **40**, and the microphone assembly **50**.

(35) Referring to FIG. 5, in some embodiments, the housing **10** may include a holder **11** and a sleeve **12** sleeved on the holder **11**. The holder **11** is configured to provide support for components such as the vaporization assembly **20**, the main control board **40**, the microphone assembly **50**, and the power source device **60**. The sleeve **12** is sleeved on the holder **11** in the axial direction, and is configured to protect components on the holder **11**, and in some embodiments, a cross section of the sleeve **12** can be in the shape of a racetrack.

(36) In some embodiments, the holder **11** may include an elongated holder body **111**, a lower end cap **112** integrally connected to a lower end of the holder body **111**, and an upper end cap **113** integrally connected to an upper end of the holder body **111**. In some embodiments, the lower end cap **112** and the upper end cap **113** may be in the shape of a racetrack-shaped flat plate, and the lower end cap **112** and the upper end cap **113** are perpendicular to a longitudinal axis of the holder body **111**. The lower end cap **112** and the upper end cap **113** are respectively blocked on a lower end opening and an upper end opening of the sleeve **12** to define a sealed cavity **110** to

accommodate electronic components such as the vaporization assembly **20**, the main control board **40**, and the microphone assembly **50**. A plurality of air inlet holes **120** configured to cause the cavity to be in communication with the outside may be provided on the sleeve **12** and at a position close to an upper end of the sleeve **12**. The air inlet holes **120** cause the vaporization assembly **20** to be in communication with the outside, so that ambient air can enter the vaporization assembly **20**.

(37) In some embodiments, the upper end cap **113** may include a vent hole **1130** running through in an up-down direction, and the vent hole **1130** is in communication with the vaporization assembly **20** to cause a mixture of the aerosol generated by the vaporization assembly **20** and air to be discharged. In some embodiments, the upper end cap **113** may include a plurality of arc-shaped buckles **1132** protruding on a lower side. The plurality of buckles **1132** encircle to cause the upper end of the vaporization assembly **20** to be rotatably connected to the plurality of buckles **1132**, so as to cause the vaporization assembly **20** to be fixed with the upper end cap **113**. In some embodiments, the upper end of the vaporization assembly **20** can be detachably screwed on the buckle **1132**, so that the upper end of vaporization assembly can be replaced when the vaporization assembly **20** is damaged.

(38) As shown in FIG. 6 to FIG. 9, in some embodiments, the vaporization assembly **20** may include a cylindrical vaporization base **21** mounted on a lower side of the upper end cap **113**, a cylindrical heat shield **22** mounted in the vaporization base **21** in the axial direction, a cylindrical heating pot **23** mounted in the heat shield **22** in the axial direction, a seal ring **24** mounted between a top wall **213** of the vaporization base **21** and the upper end of the heat shield **22** and the upper end of the heating pot **23**, and a seal ring **25** mounted on a side wall **211** of the vaporization base **21** and a temperature sensing element **26**. The vaporization base **21** is configured to carry the heat shield **22** and the heating pot **23**, and the heat shield **22** is configured to prevent the excessive heat released by the heating pot **23** from being transferred to surrounding electronic components. The heating pot **23** is configured to accommodate and heat the aerosol-generation substrate to generate the aerosol. The seal ring **24** is configured to seal and position the upper end of the heat shield **22** and the upper end of the heating pot **23**. The seal ring **25** is configured to cause the air inlet hole **2110** on an upper portion of the side wall **211** of the vaporization base **21** to be in communication with the air inlet hole **120** on the sleeve **12** of the housing **10** in a sealing and air guiding manner (as shown in FIG. 10). The temperature sensing element **26** is arranged between the heat shield **22** and the heating pot **23**, and is configured to sense a temperature of a periphery of the heating pot **23** to provide a temperature signal to a control system, preventing the temperature of the heating pot **23** from being too high or too low during operation.

(39) Referring to FIG. 7, in some embodiments, the vaporization base **21** may include a cylindrical side wall **211** made of a high temperature-resistant plastic material, a plate-shaped bottom wall **212** that is in snap-fit connection to a lower end of the side wall **211**, an annular plate-shaped top wall **213** that is integrally connected to the upper end of the side wall **211**, and a cylindrical neck **214** that is integrally connected to a top portion of the top wall **213**. The side wall **211**, the bottom wall **212**, and the top wall **213** together define a cylindrical accommodating space configured to accommodate the heat shield **22** and the heating pot **23** therein. The neck **214** surrounds a central through hole of the top wall **213** and is embedded in the vent hole **1130** of the upper end cap **113**, and is connected to the suction nozzle assembly **30** in a mechanical, air guiding, and detachable manner. The bottom wall **212** is detachably connected to the side wall **211**, to accommodate the heat shield **22** and the heating pot **23** first.

(40) In some embodiments, the vaporization base **21** may include a plurality of elongated engagement protrusions **215** with a wedge-shaped cross section, and each of the engagement protrusion **215** includes a helically inclined top surface. The engagement protrusions **215** are spaced apart in a circumferential direction of the outer wall surface of the upper end of the side wall **211**, and can be respectively rotatably connected to (similar to threaded connection) a plurality

of buckles **1132** of the upper end cap **113**, so as to implement the detachable connection between the vaporization base **21** and the upper end cap **113**.

(41) In some embodiments, an air inlet hole **2110** running through in a thickness direction may further be provided on the side wall **211** and at a position close to an upper end of the side wall **211**. In some embodiments, the vaporization base **21** may include a seal base **216** integrally formed on the outer surface of the side wall **211** and surrounding the air inlet hole **2110**. The seal base **216** defines a square accommodating cavity for the seal ring **25** to be embedded therein. In some embodiments, the side wall **211** may further include an outer protruding portion **2112** to cooperate with a protruding portion N of the heat shield **22**.

(42) In some embodiments, the vaporization base **21** may include a pair of bosses **217** respectively arranged on two opposite outer surfaces of the lower end of the side wall **211**, for the bottom wall **212** to be in snap-fit connection with the bosses. In some embodiments, the vaporization base **21** may include a pair of buckling arms **218**. The pair of buckling arms **218** are respectively arranged on two opposite sides of the bottom wall **212** and protrude upward to be respectively in snap-fit connection with the pair of bosses **217** on the side wall **211**, so as to implement snap-fit connection between the bottom wall **212** and the side wall **211**. In some embodiments, the bottom wall **212** may include a vent hole **2120** that is in communication with the microphone assembly **50** in an air guiding manner, so that the microphone assembly **50** is driven to operate when a negative pressure is generated in the vaporization base **21**. In some embodiments, the vaporization base **21** may further include a seal ring **219**, and the seal ring **219** is sandwiched between a lower end surface of the side wall **211** and a top surface of the bottom wall **212** to seal a gap therebetween.

(43) As shown in FIG. 7 and FIG. 9, in some embodiments, the heat shield **22** can be made of a ceramic material. The heat shield **22** may include a cylindrical side wall **221**, a bottom wall **222** integrally combined with the lower end of the side wall **221**, a plurality of ribs **223** that are spaced apart in parallel integrally formed on the longitudinal direction of the outer surface of the side wall **221**, and a plurality of legs **224** that are spaced apart integrally formed on a bottom surface of the bottom wall **222**. When the heat shield **22** is arranged in the vaporization base **21**, the side wall **221** and the ribs **223** together with the side wall **211** of the vaporization base **21** define an S-shaped airflow channel from top to bottom.

(44) In some embodiments, the side wall **221** may include a main body portion M with a C-shaped cross section and a protruding portion N with a U-shaped cross section that is mouth-to-mouth integrally combined with the C-shaped main body portion M. The main body portion M is configured to accommodate the heating pot **23**, and the protruding portion N is configured to accommodate the temperature sensing element **26**. The bottom wall **222** is configured to support the heating pot **23**, and may include an air inlet hole **2220** that is provided in a middle portion and runs through in a longitudinal direction and a thermal insulation protrusion **2222** formed in a middle portion of the top surface of the bottom wall **222**. The air inlet hole **2220** is configured to allow the ambient air to enter the heating pot **23**. The thermal insulation protrusion **2222** is supported on a bottom surface of the heating pot **23** to reduce a contact area between the heating pot **23** and the heat shield **22** and prevent excessive heat from being transferred to the heat shield **22**.

(45) Referring to FIG. 10, each rib **223** extends along the outer surface of the side wall **221** in a circumferential direction, and extends from one side of the protruding portion N through the main body portion M to an other side of the protruding portion N. The ribs **223** form a plurality of C-shaped air guide grooves **2230** extending in a circumferential direction on the outer surface of the side wall **221**. A notch **2232** is provided on a side of each rib **223** that is close to the protruding portion N, and the notches **2232** of the adjacent ribs **223** are close to different sides of the protruding portion N. The air guide grooves **2230** are connected end to end to form an S-shaped airflow channel from top to bottom (as shown in an arrow group W in FIG. 10). The S-shaped airflow channel can cover most areas of the outer surface of the side wall **221** of the heat shield **22**,

so that the ambient air entering from the outside can circulate therein before entering the heating pot **23**. The ambient air is in sufficient contact with the surface of the heat shield **22**, and on the one hand, the heat shield **22** can be effectively cooled to reduce the heat transferred to the outer surface of the housing **10**; and on the other hand, when the heated air enters the heating pot **23** again, the probability of occurrence of the condensation phenomenon in the heating pot **23** can be reduced.

(46) In some embodiments, based on the consideration of the overall size, the rib **223** are not arranged on the outer surface of the protruding portion N. Therefore, the air guide groove is neither provided on the outer surface of the protruding portion N, the ambient air cannot flow through the surface of the protruding portion N, and the temperature of the protruding portion N may be relatively high. However, during assembly, the protruding portion N can be arranged toward the power source device **60**, so that the power source device **60** can insulate heat, thereby preventing local heat generation at a position on the housing **10** corresponding to the protruding portion N. It may be understood that, in some embodiments, if the temperature sensing element **26** does not need to be arranged at the protruding portion N, the protruding portion N may not be arranged on the heat shield **22**, and is cylindrical as a whole. In this case, the airflow channel can be distributed in an entire circumferential direction of the heat shield **22** to implement better heat dissipation to the heat shield **22**.

(47) It can be further understood that the airflow channel on the surface of the heat shield **22** is not limited to the layout shown in the figure. An airflow channel with another layout is also applicable provided that the ambient air can conduct relatively sufficient heat exchange on the outer surface of the heat shield **22** before entering the heating pot **23**.

(48) As shown in FIG. **6**, the legs **224** are configured to support the upper surface of the bottom wall **212** of the vaporization base **21**, and gaps **2240** are formed between the adjacent legs **224**. The gaps **2240** allow the S-shaped airflow channel to be in communication with the air inlet hole **2220** on the bottom wall **222** of the heat shield **22**.

(49) As shown in FIG. **6** to FIG. **9**, in some embodiments, the heating pot **23** can be made of stainless steel, a heating circuit can be printed on the outer surface of the heating pot **23** to generate heat, and the heat is conducted into the heating pot **23**. In addition, an infrared radiation coating layer **233** can be printed on an inner side of the heating pot **23** to radiate heat to a to-be-heated medium in the pot. As shown in the figure, in some embodiments, the heating pot **23** may include a cylindrical side wall **231** and a bottom wall **232** that is integrally formed at the lower end of the side wall **231**. The bottom wall **232** is provided with a plurality of air inlet holes **2320** distributed in a net shape, and the air inlet holes **2320** are in communication with the air inlet hole **2220** of the heat shield **22**. An electrode lead **2321** may be connected to the bottom wall **232**. When the heating pot **23** is arranged in the heat shield **22**, there is a spacing **2310** between the side wall **231** and the side wall **221** of the heat shield **22**, so as to cause the air to isolate the side wall **231** and the side wall **221** of the heat shield **22**. In this way, the heat on the side wall **231** is reduced to be transferred to the side wall **221** by heat conduction.

(50) As shown in FIG. **1** to FIG. **4**, in some embodiments, the suction nozzle assembly **30** may include a suction nozzle base **31** with an air guide channel **310**, a bowl-shaped air guide member **32** protruding from a bottom portion of the suction nozzle base **31**, and a suction nozzle **33** rotatably connected to a top portion of the suction nozzle base **31**. The suction nozzle **33** includes an air guide channel **330** that is in communicated with the air guide channel **310**. In some embodiments, the air guide member **32** may include a bottom wall **321** and a side wall **322** that is integrally combined with a periphery of the bottom wall **321**. The lower end of the air guide member **32** can be embedded in the neck **214** of the vaporization base **21** to cause an inner cavity of the heating pot **23** to be in air-tight communication with the air guide channel **310** of the suction nozzle base **31**. The bottom wall **321** of the air guide member **32** is provided with a plurality of air inlet holes **3210** running through in an up-down direction. The upper end of the side wall **322** is fixed in the bottom wall of the suction nozzle base **31**.

(51) It can be understood that the foregoing technical features can be used in any combination without limitation.

(52) The foregoing descriptions are merely embodiments of the present disclosure, and the protection scope of the present disclosure is not limited thereto. All equivalent structure or process changes made according to the content of this specification and accompanying drawings in the present disclosure or by directly or indirectly applying the present disclosure in other related technical fields shall fall within the protection scope of the present disclosure.

(53) While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

(54) The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. An electronic vaporization device, comprising: a housing; a vaporization assembly arranged in the housing, the vaporization assembly comprising a vaporization base; a heat shield arranged in the vaporization base; and a heating pot arranged in the heat shield, wherein an airflow channel for ambient air to enter the heating pot is formed on an outer surface of the heat shield, wherein the vaporization base comprises a cylindrical first side wall, wherein a first air inlet hole that is in communication with the airflow channel is provided on the cylindrical first side wall and at a position close to an upper end of the cylindrical first side wall, and wherein the housing is provided with a second air inlet hole that is in communication with the first air inlet hole.
2. The electronic vaporization device of claim 1, wherein the airflow channel is distributed over most areas of the outer surface of the heat shield.
3. The electronic vaporization device of claim 1, wherein the airflow channel extends from an upper end to a lower end of the heat shield.
4. The electronic vaporization device of claim 1, wherein the airflow channel is distributed in an S shape on the outer surface of the heat shield.
5. The electronic vaporization device of claim 4, wherein the heat shield comprises a cylindrical second side wall, wherein the airflow channel comprises a plurality of C-shaped air guide grooves spaced apart in parallel in a longitudinal direction of an outer surface of the cylindrical second side wall, and wherein the air guide grooves are connected end to end.
6. The electronic vaporization device of claim 5, wherein the cylindrical second side wall comprises a main body portion with a C-shaped cross section and a protruding portion with a U-shaped cross section that is mouth-to-mouth combined with the main body portion, and wherein the

plurality of C-shaped air guide grooves are formed in a longitudinal direction of an outer surface of the main body portion.

7. The electronic vaporization device of claim 6, wherein the vaporization assembly comprises a temperature sensing element, the heating pot comprises a cylindrical third side wall, and the cylindrical third side wall is axially arranged in the cylindrical second side wall, and wherein the temperature sensing element is arranged between the cylindrical third side wall and the cylindrical second side wall and is arranged in a space defined by the protruding portion.

8. The electronic vaporization device of claim 1, wherein an outer surface of the cylindrical first side wall is provided with a seal base surrounding the first air inlet hole, wherein the vaporization assembly comprises a first seal ring embedded in the seal base, and wherein the first seal ring causes the first air inlet hole to be in communication with the second air inlet hole in a sealing and air guiding manner.

9. The electronic vaporization device of claim 1, wherein the vaporization base comprises an annular plate-shaped top wall that is integrally connected to the upper end of the cylindrical first side wall and a plate-shaped bottom wall that is in snap-fit connection to a lower end of the cylindrical first side wall, and wherein the upper end and the lower end of the heat shield respectively abut against the top wall and the bottom wall.

10. The electronic vaporization device of claim 1, wherein the heat shield comprises a cylindrical second side wall and a first bottom wall combined with a lower end of the cylindrical second side wall, and wherein an upper surface of the first bottom wall is provided with a thermal insulation protrusion configured to support the heating pot.

11. The electronic vaporization device of claim 1, wherein the heat shield comprises a cylindrical second side wall, and the heating pot comprises a cylindrical third side wall, and wherein the cylindrical third side wall is axially arranged in the cylindrical second side wall and a spacing is formed between an outer surface of the cylindrical third side wall and an inner surface of the cylindrical second side wall.

12. The electronic vaporization device of claim 1, wherein an outer surface of the heating pot is provided with a heating circuit and an inner surface of the heating pot is provided with an infrared radiation coating layer.

13. The electronic vaporization device of claim 1, further comprising: a suction nozzle assembly detachably connected to the housing and connected to the heating pot in an air guiding manner.

14. The electronic vaporization device of claim 1, wherein the vaporization assembly is detachably connected to the housing.

15. The electronic vaporization device of claim 14, wherein the vaporization base comprises a cylindrical side wall and a plurality of elongated engagement protrusions with a wedge-shaped cross section, each of the engagement protrusion comprising a helically inclined top surface, and wherein the engagement protrusions are spaced apart in a circumferential direction of an outer wall surface of the cylindrical side wall that is close to an upper end of the cylindrical side wall so as to be detachably and rotatably connected to the housing.

16. The electronic vaporization device of claim 15, wherein the housing comprises a holder, the holder comprises an upper end cap, and a plurality of arc-shaped buckles protrude from a lower side of the upper end cap, and wherein the buckles encircle to be respectively rotatably connected to the plurality of engagement protrusions.
